Utkarsh Dewangan

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GitHub Dashboard

SUMMARY

Ambitious CSE student at NHCE Bengaluru (VTU | CGPA 8.7) with a foundation in Linux, shell scripting, networking, and Git - actively pursuing a DevOps career path. Co-developing an ML-based Network Intrusion Detection System under an IEEE internship, using Random Forest, XGBoost, ANN, and SHAP explainability on real-world datasets. A filed patent and hands-on research reflect my drive to build intelligent, secure, and connected systems aligned with Nokia's vision.

EDUCATION

B.E. in Computer Science 2023 – 2027
New Horizon College of Engineering (Affiliated to Visvesvaraya Technological University) Bengaluru
CGPA: 8.7

XII Standard (Central Board of Secondary Education) 2022 – 2023
Kendriya Vidyalaya NERIST Arunachal Pradesh
CG PA: 7.5

X Standard (Central Board of Secondary Education) 2020 – 2021
Kendriya Vidyalaya NERIST Arunachal Pradesh
CGPA: 8.9

PROJECTS

Python – Expense Tracker & Stock Analysis | Python, Tkinter, SQLite, yfinance

- Built a desktop app for personal finance management with expense tracking, budget monitoring, and real-time stock market analysis.
- Implemented secure user authentication using bcrypt and designed a normalized SQLite database with full CRUD operations.
- Integrated Yahoo Finance API for live stock data, technical indicators (SMA, EMA, RSI, MACD), and a Linear Regression-based price prediction module.

PROFESSIONAL EXPERIENCE

Multi-Attack Network Intrusion Detection System (NIDS) Python, Scikit-learn, XGBoost, TensorFlow/Keras, SHAP, Pandas, NumPy | IEEE IamPro 2026 Internship (Ongoing)

- Built a multi-class NIDS classifying 14+ attack types (DoS, DDoS, Botnet, Brute Force) across CICIDS2017 & UNSW-NB15 datasets (~2.8M+ records) using Random Forest, XGBoost, and ANN.
- Applied SMOTE-based class balancing, correlation-based feature selection, and SHAP explainability for interpretable, SOC-ready predictions.

PATENTS

An Integrated Fog Collection System with Multi-Source Energy Harvesting | India India
Application No. 202641005295 A | Filed: 19/01/2026 | Published: 30/01/2026

- Co-invented a fog collection system using hydrophobic mesh, piezoelectric transducers, and solar/wind energy harvesting — producing 9,000–12,000 L of potable water and 700–900 kWh annually, ~40% more efficient than conventional methods.

SKILLS

- Coding Languages: C++, Python, FastAPI
- Networking & OS: Linux, Networking fundamentals,
- Tools: Git, GitHub, Shell Scripting (Bash), Docker
- Cloud: AWS
- Databases: SQLite

CERTIFICATES

- Symposium on Data for Public Good, Indian Institute of Science (IISc)
- DBMS Course – Master the Fundamentals and Advanced Concepts, Scaler

AWARDS

Runner-Up – VIBETHON | IEEE Computer Society NHCE Student Branch, Bengaluru